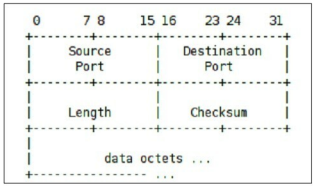

Figure 4: The UDP header *(Source: RFC 768)*

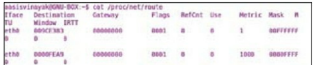

Figure 5: The */proc/net/route* file

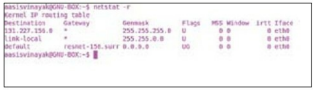

Figure 6: The output for *netstat -r*

```
      inet addr:127.0.0.1  Mask:255.0.0.0
      inet6 addr: ::1/128 Scope:Host
      UP LOOPBACK RUNNING  MTU:16436  Metric:1
      RX packets:974 errors:0 dropped:0 overruns:0 frame:0
      TX packets:974 errors:0 dropped:0 overruns:0 carrier:0
      collisions:0 txqueuelen:0
      RX bytes:153795 (153.7 KB)  TX bytes:153795 (153.7 KB)

pan0      Link encap:Ethernet  HWaddr 66:a2:20:ba:f3:10
      BROADCAST MULTICAST  MTU:1500  Metric:1
      RX packets:0 errors:0 dropped:0 overruns:0 frame:0
      TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
      collisions:0 txqueuelen:0
      RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

vboxnet0  Link encap:Ethernet  HWaddr 0a:00:27:00:00:00
      BROADCAST MULTICAST  MTU:1500  Metric:1
      RX packets:0 errors:0 dropped:0 overruns:0 frame:0
      TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
      collisions:0 txqueuelen:1000
      RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

You may use the ***down*** option to shut down an interface, and ***arp*** to enable (or disable) the use of the ARP protocol on an interface. ***io_addr addr*** is another option to start an address in the I/O space. And if you wish to assign an IP address, you may just employ the ***address*** option.

Displaying the output (content) of */proc/interrupts* and analysing it is quite handy in some cases (say, if you want to look at the number of interrupts per IRQ). You can do this by issuing the following commands:

```
aasisvinayak@GNU-BOX:~$ sudo cat /proc/interrupts
[sudo] password for aasisvinayak:
           CPU0       CPU1
  0:    2275331    1552933   IO-APIC-edge      timer
  1:        917        457   IO-APIC-edge      i8042
  8:          0          1   IO-APIC-edge      rtc0
  9:          9          4   IO-APIC-fasteoi   acpi
 12:      79030      76412   IO-APIC-edge      i8042
 17:       4193       2697   IO-APIC-fasteoi   eth2
 20:        457        159   IO-APIC-fasteoi   ehci_hcd:usb2, uhci_hcd:usb3,
uhci_hcd:usb6
 21:      83736      41592   IO-APIC-fasteoi   uhci_hcd:usb4, uhci_hcd:usb7,
HDA Intel
 22:         27         13   IO-APIC-fasteoi   ehci_hcd:usb1, uhci_hcd:usb5,
uhci_hcd:usb8
 28:      79298      55301   PCI-MSI-edge      ahci
 29:     299499     306516   PCI-MSI-edge      eth0
 30:     181527     188303   PCI-MSI-edge      i915@pci:0000:00:02.0
NMI:          0          0   Non-maskable interrupts
LOC:    1551082    2026604   Local timer interrupts
SPU:          0          0   Spurious interrupts
CNT:          0          0   Performance counter interrupts
PND:          0          0   Performance pending work
RES:     789579     767812   Rescheduling interrupts
CAL:        156        160   Function call interrupts
TLB:       5216       7330   TLB shootdowns
TRM:          0          0   Thermal event interrupts
THR:          0          0   Threshold APIC interrupts
MCE:          0          0   Machine check exceptions
MCP:         34         34   Machine check polls
ERR:          0
MIS:
```

If you wish to look at the static routing table, you can *cat* the */proc/net/route* file. Figure 5 shows the output on my PC.

You might have tried the ***netstat*** command as well. But the above commands will work even if you don't have the ***netstat*** utility. (I have seen some changes in 2.4 and 2.6, but you need not worry about this now.)

Since most of the ***distros*** have this utility, you can issue ***netstat -r*** (or ***-nr***) to obtain the routing table—readability is better in this case (refer to Figure 6). You can also use ***route -n*** for the same purpose.

/etc/services is a very vital file that helps you find how the port numbers are linked to the named services. For your reference, here is a standard entry in the file:

```
tcpmux          1/tcp                   # TCP port service multiplexer
```